

QUAD PACKING

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ABSTRACT. *Quads* or *EvenQuads* is a pattern-recognition game similar to *Set*. The goal of the game is to find “quads”, which are sets of 4 cards that satisfy a particular pattern. In this talk, we will discuss our research into the maximum number of quads in a k card layout. Specifically, we will demonstrate an upper bound for the number of quads in a k card for all k and prove this bound is tight for when k is a power of two. Furthermore, we will share the results for $4 \leq k \leq 6$.

1. INTRODUCTION

Quads or *EvenQuads* [5] is a pattern-recognition game similar to *Set* [3], where we are given a set of cards, each card has d *properties*, and each property has four *attributes*, though it also is possible to work with two attributes for some properties. The goal of the game is to find *quads*, which are subsets of four cards that satisfy a particular pattern. We will elaborate on the definition of a quad in Section 2. In this project, we will discuss our research into the maximum number of (not necessarily disjoint) quads in a set of k cards.

To preface, we will introduce the perspective of each Quads card with d properties as an element of $\mathbb{Z}_2^{2d}$. In Section 3, we will elaborate on why this perspective makes sense and introduce the Fundamental Theorem of Quads. In Section 4, we will also introduce the titular quad packing problem statement. In Section 5, we will demonstrate an upper bound for the number of quads in a set of k cards for all $k \in \mathbb{N}$. Furthermore, we will share our results for $4 \leq k \leq 7$. In Section 6, we will state five conjectures related to our findings from Section 5, their relevance to this problem, and progress on each of these conjectures, if any. In Section 7, we will describe an original computer program in C++ that allows us to determine (albeit through high time complexity, which may be improved) the value of $q(k, 2d)$, and we will share some discoveries we made upon tuning the parameters k and d .

2. THE GAME OF QUADS

Before we introduce the titular problem we will discuss in this paper, we will first explain the rules of *Quads* [4].

The game *Quads* is a card game, typically played with a deck of 64 cards, though a deck with the number of cards equal to any power of 2 will also work. Each card can be described as a set of some number of *properties*, such as shape, color, and number. In the case where the deck has 64 cards, there are 3 properties because $4^3 = 64$. In addition, each property has four *attributes*. In other words, if we suppose that the three properties of a card are shape, color, and number, then there are four possible shapes, four possible colors, and four possible numbers. Although the number of properties in a Quads deck may differ, the number of attributes per property should be exactly four since that is a key characteristic of *Quads*. For example, the card with three blue circles is shown below.

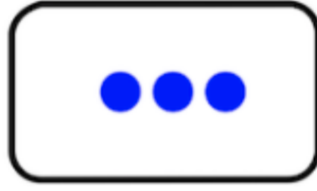


Figure 1. A *Quads* card with three blue circles [4].

Below is a figure of all 64 cards in a standard deck.

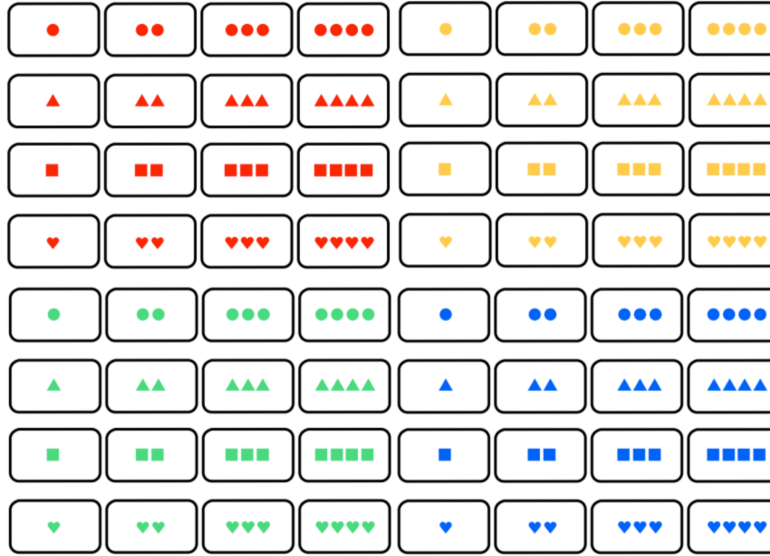


Figure 2. All 64 *Quads* cards [4].

We will now go over the rules of the game. First, we must define what a quad is.

Definition 2.1 (Quad). A set of four cards is a *quad* if for every property, either all four cards have the same attribute, all four cards have different attributes, or two cards have one attribute and the other cards have a second attribute, of that property.

We shall go over two examples of four cards that form a quad and one example of four cards that do not form a quad [4].

The four cards below form a quad because the colors are all the same (red), the shapes are all different (circle, triangle, square, and heart), and the numbers of shapes on them are the same (one).



Figure 3. An example of four cards forming a quad [4].

The four cards below form a quad because the colors are half-red and half-blue, the shapes are all the same (circle), and the numbers of shapes on them are all different (one, two, three, and four).



Figure 4. A second example of four cards forming a quad [4].

The four cards below, however, do **not** form a quad because although the colors are all different (yellow, red, green, and blue), and the numbers of shapes on them are all different (one, two, three, and four), there are two cards with circles, one card with hearts, and one card with triangles, which is not allowed.



Figure 5. An example of four cards *not* forming a quad [4].

3. FUNDAMENTAL THEOREM OF QUADS

With the rules of the game and the definition of a quad outlined, we will now introduce a new way to view a card: we may express each of the four attributes of a property with a unique element from $\mathbb{Z}_2^2$ (namely 00, 01, 10, and 11), and if there are d properties, then we may concatenate these elements into one binary encoding of length $2d$. This means that we may express every card as an element of $\mathbb{Z}_2^{2d}$, and since there are $2^{2d} = 4^d$ elements of the vector space $\mathbb{Z}_2^{2d}$, which is also the number of cards in a full *Quads* deck with d properties, there exists a bijection between the set of all cards with d properties and $\mathbb{Z}_2^{2d}$.

This representation of the set of cards will prove useful when defining key properties of a quad and will allow us to prove some more difficult results, some of which we will elaborate upon further in Section 5.

Next, we will prove a very important and fundamental theorem describing a relationship between the cards in a quad, aptly named the Fundamental Theorem of Quads, by leveraging the binary representation of *Quads* cards. Before tackling this theorem, we will state and prove the following claim.

Claim 3.1. *Let $w, x, y, z \in \mathbb{Z}_2^n$ be distinct. Then $\{w, x, y, z\}$ forms a quad if and only if $a+b+c+d = 0 \pmod{4}$.*

Proof. Suppose the cards $w, x, y, z \in \mathbb{Z}_2^n$ form a quad, for some $n \in \mathbb{N}$. Consider a specific property represented by the corresponding two bits of these elements. Among the four cards, this attribute must either be all the same, all different, or two and two. Thus, the corresponding two bits of w, x, y, z must either be all the same, all different, or two and two. If they are all the same two bits, which we will say is $a \in \mathbb{Z}_2^2$, then we have $a + a + a + a = (0, 0)$. If they are all different, then they are all the elements of $\mathbb{Z}_2^2$, so we have $(0, 0) + (0, 1) + (1, 0) + (1, 1) = (0, 0)$. Finally, if they are two and two, which we will say are $a, b \in \mathbb{Z}_2^2$, where $a \neq b$, then we have $a + a + b + b = (0, 0)$. In all three cases, if we add these two bits of each of these elements, then we have $(0, 0)$. Since this holds for every property, we will obtain $w + x + y + z = \underbrace{(0, \dots, 0)}_{n \text{ zeroes}}$.

Conversely, suppose for the sake of contradiction that $\{w, x, y, z\}$ does not form a quad, i.e., that some property of the cards breaks the condition for this to form a quad. The elements of $\mathbb{Z}_2^2$ corresponding to this property among w, x, y, z must either be of the form a, a, b, c or a, a, a, b ,

where $a, b, c \in \mathbb{Z}_2^2$ are pairwise distinct. Then we must have that $a + a + b + c = b + c \neq (0, 0)$ and $a + a + a + b = a + b \neq (0, 0)$, which means that $w + x + y + z \neq \underbrace{0 \dots 0}_{n \text{ zeroes}}$. $\square$

Armed with this claim, we are now ready to detail the statement and proof of the Fundamental Theorem of Quads.

Theorem 3.2 (Fundamental Theorem of Quads). *Let w, x, y be distinct elements of $\mathbb{Z}_2^n$. Then there is a unique element $z \in \mathbb{Z}_2^n$ such that $\{w, x, y, z\}$ is a quad [4].*

Proof. We see that

$$w + x + y + (w + x + y) = 0,$$

which means that $\{w, x, y, w + x + y\}$ forms a quad. Now, given a quad $\{w, x, y, z\}$, we know that $w + x + y + z = 0$ from the previous claim, which means that $z = w + x + y$, which is clearly unique. $\square$

This simple yet important theorem will provide the baseline for the problem we wish to tackle in Section 4 and also give us some intuition on how we could go about proving more difficult results.

To conclude this section, we will define the notion of *completing* a quad given three distinct cards with a fourth one.

Definition 3.3 (Quad Completeness). A card z *completes* a set of three cards $\{w, x, y\}$ if $\{w, x, y, z\}$ is a quad. Furthermore, given a subset of three cards $\{w, x, y\}$ of some deck of cards, we say that this subset is *complete* if $z = w + x + y$ is also in this deck of cards.

By the Fundamental Theorem of Quads, this fourth card is guaranteed to exist and is unique.

4. QUAD PACKING

With preliminary definitions and the Fundamental Theorem of Quads in our hands, we are now ready to introduce the research problem of our interest: Given any $d \in \mathbb{N}$ and $4 \leq k \leq 2^{2d}$, what is the maximum number of quads that can exist in a deck of k cards with each one having d properties?

Though briefly hinted at in Section 1, we will reintroduce all definitions relevant to the Maximum Quad Packing problem.

Definition 4.1 ($q(k, d)$). For any $d \in \mathbb{N}$ and $4 \leq k \leq 2^{2d}$, the maximum number of quads that can exist in a deck of k cards with each one having d properties is denoted by $q(k, 2d)$.

5. FINDINGS

With the definitions from Section 4, we will now go over two results that we were able to prove in our project.

Theorem 5.1 (Upper Bound on $q(k, d)$). *Given any set of k cards in $\mathbb{Z}_2^{2d}$, for some $4 \leq k \leq 2^{2d}$, $\frac{\binom{k}{3}}{4}$ is an upper bound for the number of quads in the set.*

Proof. Let S be any set of k cards in $\mathbb{Z}_2^{2d}$, for some $4 \leq k \leq 4^d$. To preface, we will call a subset of 3 cards $\{w, x, y\} \subseteq S$ *complete* if $w + x + y \in S$ and *incomplete* if $w + x + y \notin S$. Let c be the number of complete triples and i be the number of incomplete triples. Then $\binom{k}{3} = c + i$. Let Q

be the total number of quads in S . For each quad, there are $\binom{4}{3} = 4$ complete triples, so by the Fundamental Theorem of Quads, $c = 4Q$. We then have that

$$\frac{\binom{k}{3}}{4} = \frac{c}{4} + \frac{i}{4} = \frac{4Q}{4} + \frac{i}{4} = Q + \frac{i}{4}.$$

Since Q, i, k are non-negative, we have that $Q \leq \frac{\binom{k}{3}}{4}$. $\square$

In addition to the result above, we have also explicitly found $q(k, d)$ for all $4 \leq k \leq 7$. Henceforth, assume that for each expression, d is chosen such that $2^{2d} \geq k$.

Theorem 5.2 (Specific Results for $4 \leq k \leq 7$). *The following is true: $q(4, d) = 1$, $q(5, d) = 1$, $q(6, d) = 3$, $q(7, d) = 7$.*

Proof. We will prove each expression individually.

If $k = 4$, then trivially, there can be at most 1 quad because we may choose a set of $k = 4$ cards that themselves form a quad, and we cannot form more than 1 quad from a set of 4 cards because there is exactly one subset of 4 cards. Hence, $q(4, d) = 1$.

If $k = 5$, then we will show that there can be at most 1 quad; by extension, $q(5, d) = 1$. Suppose we have 5 cards, a, b, c, d , and e , and suppose, without loss of generality, that a, b, c , and d form a quad. By the fundamental theorem of quads, three cards cannot be in more than one quad. Any set of 4 cards must contain at least 3 of a, b, c , and d , and thus cannot be a quad unless the set contains exactly a, b, c , and d . Therefore, a, b, c , and d can be the only quad. Since this was an arbitrary set of cards with at least one quad, hence there cannot be more than one quad in any set of 5 cards.

We will show that there can be exactly 0, 1, or 3 quads in a set of 6 cards; by extension, $q(6, d) = 3$. Let a, b, c, d, e , and f be our cards.

It has been shown that we are not guaranteed a quad until we lay out 10 cards [2]. Thus, it is possible to have a set of 6 cards with 0 quads.

For the case of 1 quad, we can assume without loss of generality that $\{a, b, c, d\}$ forms a quad. Now, we simply choose any card e and choose f such that f does not complete a quad with any pair from $\{a, b, c, d\}$ together with e as the third card.

For the case of 3 quads, we can assume without loss of generality that $\{a, b, c, d\}$ forms a quad. By the Fundamental Theorem of Quads, any new quad cannot contain three of a, b, c, d , so it must contain e and f . Without loss of generality, assume $\{a, b, e, f\}$ is a quad. We then have $a + b + c + d = 0$ and $a + b + e + f = 0$. Therefore, $c + d + e + f = 0$, making our third quad $\{c, d, e, f\}$.

As previously established, if we were going to add an additional quad to our set up with 3 quads, it must include e and f . However, choosing any of a, b, c, d as our third card forces one of the pre-existing quads. Therefore, $q(6, d) = 3$.

We will show there can be exactly 0, 1, 2, 3, or 7 quads in a set of 7 cards; by extension, $q(7, d) = 7$. Let a, b, c, d, e, f, g be our cards.

Akin to our reasoning for there potentially being 0 quads when $k = 6$, it is possible for there to be 0 quads when $k = 7$.

For the case of 1 quad, we may take a set of 6 cards with one total quad and add another card that does not complete any quads.

For the case of 2 quads, we may take a set of 6 cards with 2 quads with one card overlapping between them and add another card that does not complete any quads.

For the case of 3 quads, we may take a set of 6 cards with 3 total quads and add another card that does not complete any quads.

For the case of 7 quads, we may take a set of 6 cards with 3 total quads (WLOG $\{a, b, c, d\}$, $\{a, b, e, f\}$, and $\{c, d, e, f\}$) and add another card g that completes $\{a, c, e\}$. We then form the new quads $\{a, c, e, g\}$, $\{a, d, f, g\}$, $\{b, c, f, g\}$, and $\{b, d, e, g\}$, and showing that these are indeed quads is left as an exercise to the reader. $\square$

6. CONJECTURES

We made the following five conjectures:

- $q(k, d) = q(k + 1, d)$.
- $q(2^n + 1, d) = q(2^n, d)$.
- $q(2^n, d) = q(2^n - 1, d) + \frac{\binom{2^n - 1}{2}}{3}$.
- Let S be a set of k cards with $q(k, d)$ quads. There exists a card $a \notin S$ such that $S \cup \{a\}$ has $q(k + 1, d)$ quads.
- For any integer n such that $2 \leq n \leq 2d$, there exists a set of $k = 2^n$ cards in $\mathbb{Z}_2^{2d}$ that has $\frac{\binom{k}{3}}{4}$ quads.

The first conjecture was thought of after experimenting with the C++ program that seeks to find $q(k, 2d)$ for variables k and d , as we will further explain in Section 7. The second conjecture was thought of after realizing some of the values that the C++ program outputted, which we will reveal in Section 7. The third conjecture was thought of both due to the C++ program, namely $k = 15$ and $k = 16$, and our results for $k = 7$ and $k = 8$ from our findings from Section 5. Finally, if the fourth conjecture were true, then it would be significantly easier to find an explicit construction to build maximal sets for any arbitrary integer $k \geq 4$.

The fifth conjecture was based on the following incorrect proof:

Proof. Let n be an integer such that $2 \leq n \leq 2d$. Let $x_1, \dots, x_n$ be distinct non-zero vectors in $\mathbb{Z}_2^{2d}$. Then the span of $x_1, \dots, x_n$ contains 2^n vectors. Let a, b, c be in the span of $x_1, \dots, x_n$. Then $a + b + c$ is in the span of $x_1, \dots, x_n$. Furthermore, $a + b + c$ is distinct from a, b, c by the Fundamental Theorem of Quads. Therefore, all triples in the set are completed, and so the number of quads in the set is $\frac{\binom{k}{3}}{4}$. $\square$

Alas, the span of a set of arbitrary nonzero vectors $\{x_1, \dots, x_n\} \subseteq \mathbb{Z}_2^{2d}$ does not necessarily have cardinality 2^n , with the following example with $n = 3$ and $d = 2$: $x_1 = 1010$, $x_2 = 1110$, $x_3 = 0110$. Here $|\text{Span}\{x_1, x_2, x_3\}| = 6$. Thus, more work will have to be done. However, this upper bound is exact for $k = 4$, as proven in Section 5, and the computer program in Section 7 confirms that the upper bound is indeed exact for $k = 8, 16$.

7. CODE

In hopes of discovering new insights that may help advance progress on the Maximum Quads Problem, we proposed a computer program in C++ that finds $q(k, 2d)$ through a brute-force algorithm. The GitHub repository that stores the code for this computer program is cited here and may be found in References [1].

The algorithm iterates through all $\binom{2^{2d}}{k}$ subsets of size k from a deck of 2^{2d} cards, and for each subset, we iterate through all subsets of four cards $\{w, x, y, z\}$ chosen from this specific subset of k cards in dimension d . For each quad $\{w, x, y, z\}$, we check whether $a \oplus b \oplus c \oplus d = 0$; i.e., whether the bitwise exclusive-or of w, x, y, z is equal to 0. If it is, then $\{w, x, y, z\}$ forms a quad, and so we increment our count by 1. Otherwise, we do not. Upon completion, this algorithm will return and print out the value of $q(k, 2d)$.

When we ran this code with $d = 2$ (hence 16 total cards), we obtained the following results.

k	$q(k, 2)$
9	14
10	18
11	26
12	39
13	55
14	77
15	105
16	140

Most notably, $q(9, 2) = q(8, 2)$, which led us to make the third conjecture in Section 6, and $q(16, 2) = 140 = \frac{\binom{16}{3}}{4}$, which corroborates with the second theorem in Section 5.

Another key observation was that after running the code by fixing k to be small ($4 \leq k \leq 8$) and ranging d from 3 to 6, inclusive, we found that the outputs were the same. This observation made us conjecture the first conjecture listed in Section 6. If this conjecture is true, then we would be able to choose the smallest d such that $4^d \geq k$, which would mean that less time would be necessary to verify results.

One goal that we have for this algorithm is making it polynomial-time with respect to both k and d . Our current algorithm iterates through all $\binom{2^{2d}}{k}$ subsets of k cards chosen from a deck of size 2^{2d} , but if it is possible to reduce the amount of checking (e.g., finding a way to partition the deck of cards into smaller categories and only testing one of them), then we hope to do so as an exponential-time algorithm is very unideal. One way to make this polynomial-time is by being able to prove the conjecture that d does not matter. By doing this, we may take 2^{2d} to be around the size of k , and the runtime would be polynomial with respect to k (and d).

8. CONCLUDING REMARKS

The game of *Quads* is a simple yet mathematically deep card game. Through the use of computer programming and mathematical proofs leveraging the unexpected perspective of these cards as elements of an abstract vector space, we have paved the way for this card game to be solved. In the future, we hope to both prove or disprove at least one of the four conjectures from Section 6 and improve the time complexity of the algorithm from Section 7, potentially using some of these conjectures, if true, to our advantage. However, with definitive results for k a power of two and a computer program that will eventually output the correct value of $q(k, d)$, we should not be very far off from discovering new grounds with this game.

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